

Detailed Installation Guide — Apache Spark with Scala (Windows)

STEP 1 — INSTALL JAVA (VERY IMPORTANT)

Spark requires Java to run.

1. Open Browser

Search:

Temurin JDK 8 download

2. Open Official Website

Choose:

Eclipse Temurin

3. Select These Options

Option	Select
Operating System	Windows
Architecture	x64
Package Type	JDK
Version	8

4. Download the `.msi` Installer

Example:

OpenJDK8U-jdk_x64_windows_hotspot.msi

5. Install Normally

Just click:

Next → Next → Finish

6. VERIFY JAVA

Open:

Command Prompt

Run:

```
java -version
```

Expected Output:

```
java version "1.8..."
```

STEP 2 — INSTALL SCALA

1. Open Browser

Search:

Scala download

2. Open Official Website

scala-lang.org

3. Click

Download Scala

4. Download Windows Installer

Example:

```
scala-2.13.x.msi
```

5. Install Normally

Click:

```
Next → Next → Finish
```

6. VERIFY SCALA

Open Command Prompt:

```
scala -version
```

Expected Output:

```
Scala code runner version 2.x
```

STEP 3 — INSTALL APACHE SPARK

1. Open Browser

Search:

```
Apache Spark download
```

2. Open Official Website

```
spark.apache.org
```

3. Click

Download Spark

4. IMPORTANT OPTIONS

Choose:

Option	Select
Spark version	latest stable
Package type	Pre-built for Apache Hadoop

5. Click Download

Example downloaded file:

spark-3.x-bin-hadoop3.tgz

STEP 4 — EXTRACT SPARK

Use:

- WinRAR OR
- 7-Zip

Extract To Example Location

D:\Spark

After extraction:

D:\Spark\spark-3.x-bin-hadoop3

STEP 5 — SET ENVIRONMENT VARIABLES

1. Search in Windows

Environment Variables

Open:

Edit the system environment variables

2. Click

Environment Variables

3. Under System Variables

Find:

Path

Click:

Edit

4. Add SCALA PATH

Example:

C:\Program Files (x86)\scala\bin

5. Add SPARK PATH

Example:

D:\Spark\spark-3.x-bin-hadoop3\bin

6. Click OK Everywhere

STEP 6 — VERIFY SPARK

Open Command Prompt:

```
spark-shell
```

IF SUCCESSFUL

You will see:

```
Spark welcome message
```

and:

```
scala>
```

prompt.

IF `spark-shell` IS NOT FOUND

Restart:

- Command Prompt OR
- PC

then try again.

STEP 7 — RUN SIMPLE SCALA PROGRAM

Inside:

```
spark-shell
```

Type:

```
println("Hello Spark")
```

OUTPUT

```
Hello Spark
```

DONE

You now have:

- Java installed
 - Scala installed
 - Apache Spark installed
 - Spark shell working
-

IMPORTANT UNDERSTANDING

For SPPU practicals:

Usually running:

```
spark-shell
```

and executing small Scala commands is enough.

You are generally expected to:

- understand Spark basics
- run simple programs
- explain concepts
- answer viva questions

and not build large enterprise applications.